TECHNICAL REQUIREMENTS SHEET· NIHVA • RICE PACKAGING MACHINE			
SR NO	**ITEMS**	**E L Wiegh**	**NIHVA Remark**
1	CONTROL VALVE	SS316	SS316 OK
2	WEIGHING HOPPER	SS316	SS316 OK
3	DISCHARGE HOPPER	SS316	SS316 OK
4	BAG CLAMPING PNEUMATICS	ALUMINIUM	ALUMINIUM OK
s	Load Cells	Alluminium	Alluminium OK
6	SUPPORT STRUCTURE	MS POWDER COATED	MS POWDER COATED OK
7	2 NEEDLE 4 THREAD STTITCHING MACHINE	MS - GABBAR MAKE	OK
8	Belt make	DERCO	OK
9	Belt type	PVC BELT food grade	OK
10	Gear motor	Bonfiglioli	OK
11	Idler Rollers	E L Wiegh	OK
12	Transmission Chain	Tl -Diamond	OK
13	Transmission 'V' Belts	Fenner/Gates	OK
14	Drive & Tail Pulley	Should be Crowned	OK
15	Bearing/Pillow Blocks	SKF/NTN/FAG	SKF Food Grade
16	Modular Belts	Eurobelt/Intralox	Intralox
17	Belt conveyor rollers	MS plated	OK
18	Aux. Relay/24V DC relay	Siemens	Sie mens only
19	Power Cables	Finolex, Polycab, Lapp	Polycab
20	Control Panel, Boxes, Consoles	Local make MS powder coated	OK

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21	MCB & MCCB	SCHNEIDER	Siemens only
22	Over Load Relay	SCHNEIDER	Siemens only
23	Power Contractor	SCHNEIDER	Siemens only
24	MCCB/ Main incomer	SCHNEIDER	Siemens only
25	Terminals	COONECTWELL	OK
26	Pushbuttons illuminated	Tecknic	OK
27	Emergency Push button	Tecknic	OK
28	16 port Ethernet switch	D-Link	OK
29	Profibus cable	Siemens	Siemens only
30	Ethernet cable	Finolex / Polycab / Lapp	Polycab
31	PLC	Siemens	Siemens only
32	Sensors	P&F	OK
33	Load Cells	Thames side i.e Thames	OK
34	Actuators	Techno/SMC/FESTO	SMC/Festo only
35	Control Panel Accessories	Schneider	Siemens only
36	Belt type	PVC BELT food grade	Jointless Belt PVC Food Gradeis required
37	Finishig	Mirror/Matte	OK
38	Main Units of the machine	Simplex Bagging Machine, Stitching Conveyor, Telescopic Stitching stand & Heavy duty gabbar machine, Bag Gripper Unit, Printer Stand	OK
Additional Note: 'E L Weigh' will be solely responsible for the performance of the machine with the changed parameters like height, internal design, pneumatic slide **gates,** etc. of the machine from your end.			

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General Documentation and Regulatory Requirements

A. STATUTORY/ REGULATORY REQUIREMENT & STANDARD

1. The process equipment design, fabrication, assembly, installation and operation shall be in accordance with cGMP standards and shall be equipped with all the necessary documentation files for FDA/FSSAI and Indian regulatory authority inspection suitable for the Pharmaceutical and Natural Product industry.

2. Pressure based equipment's shall conform to ASME and Indian regulatory requirements

3. Final product equipment's shall comply with ASME BPE 2016

4. All elastomers & gaskets shall be made up of FDA/FSSAI approved material.

5. The design must be hygienic and suitable for cleaning, maintenance & operationally Feasible.

B. DOCUMENTATION

Supplier shall provide the following documentation in Soft Copy & Hard Copy whichever is applicable tothe equipment:

1. GA Drawings in AutoCAD, PDF

2. Design Specifications

3. FAT/SAT Protocols

4. As built of equipment drawings with technical specifications (Material, surface finish, wall thickness, pressure rating, etc.) in DWG format

5. General manufacturing and maintenance file

6. Pressure and temperature profile test

7. Material certificates, Food grade certificates in Vendor & Sub Vendor's letter head, MOC be suitable forCIP, Caustic Chemicals, etc., Food Contact Surface List

8. Shelf Life Certificate of material, Filters' Certificates along with rating & shelf life, cleaning procedure of filters

9. COA Report & MSDS of lubricants

10. Calibration certificates for Pressure, Temperature, proximity sensors/ probes/ gauges

11. Passivation report

12. Welder Qualifications

13. Weld standard (ASTM etc.)

14. Electric motor catalogue

15. Recommended spare parts list

16. List of bought out items

17. Maintenance & Operation manual, Troubleshoot manuals, Sub vendor's machine manuals,maintenance checklists, Lubrication checklists and details,

18. Utility details

19. Safety precautions during installing, handling & transit

20. Prohibited chemicals & maximum allowed concentrations of usable chemicals, with NSF Certificate

21. No of hours of continuous/ intermittent working per day

22. SOP for operating the all equipment

23. Particulars/ Testing parameters: Machine Units/ Parts, Speed, Power Consumption,

Maximum Operating Pressure, Operating temperature, Mass Flow Rate, Maximum Current,

Nominal Input Voltage, Dimensions, etc.

24. At the time of FAT, supplier shall be ready with an engineering file which includes all the above mentioned documents and certificates/ reports for piping, complete spray dryer with chamber, cyclone separator, coupled fluid bed dryer etc. with utility det ails , electr ical loads , piping connections, passivation, MOC, flushing, electro-polishing, welding, calibration, etc.

25. Installation/Commissioning/Handover/Validation/Testing Report etc. are required.

26. Validation report Design Vs Actual

27. Software Validation, DQ, IQ, OQ, PQ, etc.

28. . Day to Day installation & commissioning activities & Training Forms are to be filled

C. INSPECTION AND TESTING

1. Component Verification and Dimension verification as per approved GA drawing
2. Nozzle size and orientation verification
3. MOC test (Molybdenum test)
4. Pressure and temperature profile test (if any) with spray or atomizer testing.
5. Factory Acceptance Tests will be done.
6. Verification of all test and calibration certification of all instruments used during FAT.
7. Ra Test for surface measurement Capacity verification

D. SCOPE OF SUPPLY

1. The scope of supply shall include the following, comprising of all the components described in thisspecifications:
2. Design
3. Manufacturing
4. Supply
5. Supervision of installation
6. Commissioning
7. Qualification documents for equipment installed in clean/production and analytical rooms
8. Post installation support
9. The transportation shall be seal packed, following all safety and common practices to avoid undesirable Consequences.
10. Equipment shall be packed and sealed in such a way as to minimize the possibility of:

a. Mechanical damage from loading and unloading,

b. Surface scratching or denting on the outside of the equipment,

c. Intrusion of foreign external bodies into the equipment's interior

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